

# Text Feature Engineering Assignment (Real-world Dataset)

## Problem Statement

You are building a Text Processing Pipeline for a company that wants to analyze real user-generated text data such as reviews and comments, and convert them into numerical features for machine learning models. Your task is to collect real-world data and implement One Hot Encoding, Bag of Words, and TF-IDF.

## Dataset Collection (Real-world)

- 1 Scrape product reviews from Flipkart or Amazon (minimum 100 reviews)
- 2 You can use Selenium, BeautifulSoup, or any reliable scraping method
- 3 Store data in CSV format with at least one column: review\_text
- 4 Ensure data is clean and readable
- 5 Do not scrape restricted or sensitive content

## Task 1: Preprocessing

- 1 Convert text to lowercase
- 2 Tokenization
- 3 Remove punctuation
- 4 Optional: Remove stopwords (and, a, the)
- 5 Optional: Lemmatization

## Task 2: Vocabulary Creation

Build vocabulary manually or using sklearn. Print vocabulary size and analyze top frequent words.

## Task 3: Feature Engineering

- 1 One Hot Encoding (document-level vector)
- 2 Bag of Words using CountVectorizer
- 3 TF-IDF using TfidfVectorizer

## Task 4: Comparison Analysis

Create a comparison table for OHE, BoW, and TF-IDF. Explain which words are most important in TF-IDF and why common words receive lower weight.

## **Task 5: Sparse Matrix Analysis**

Print shape of matrices, calculate sparsity (percentage of zeros), and explain why sparse matrices are inefficient for large-scale systems.

## **Task 6: Real-world Questions**

- 1 Why Bag of Words fails in understanding semantic meaning (example: similar meaning words)
- 2 When to use Bag of Words and TF-IDF in industry
- 3 Limitations of TF-IDF in real applications

## **Task 7: Mini Use Case**

- 1 Perform sentiment classification (positive vs negative reviews)
- 2 Use Logistic Regression or Naive Bayes
- 3 Compare performance using BoW and TF-IDF features

## **Deliverables**

- 1 Python notebook (.ipynb)
- 2 Clean and modular code
- 3 Scraped dataset (CSV)
- 4 Output screenshots
- 5 Short report (1-2 pages) with observations and conclusions